

Review and Expanding on Various Topics

- I) Equations with no solutions or infinite solutions.
- II) Systems of equations with no solutions or infinite solutions.
- III) Graphical representations of systems of equations.
- IV) Inequalities.
- V) Graphing linear inequalities.
- VI) Graphing systems of linear inequalities.

Equations with no solutions or infinite solutions.

Determine if each of the following equations has one solution, no solution, or infinite solutions. Demonstrate algebraically.

① $-5(3x+1) = x-8$

$$-15x - 5 = x - 8$$

$$15x - 15x - 5 = x - 8 + 15x$$

$$-5 = 16x - 8$$

$$8 - 5 = 16x - 8 + 8$$

$$3 = 16x$$

$$\frac{3}{16} = \frac{16x}{16}$$

$$\frac{3}{16} = x$$

One Solution

② $3x + 7 = 2(x - 4)$

$$3x + 7 = 2x - 8$$

$$x = -15$$

One Solution

③ $2x + 3 = x + 8 + x$

$$2x + 3 = 2x + 8$$

$$3 = 8$$

No Solution

④ $-2x - 5x = 10 - 7x$

$$-7x = 10 - 7x$$

$$0 = 10$$

No Solution

$$\textcircled{5} -3(2x-3)+10x=1+4(2-3x)+16x$$

$$-6x+9+10x=1+8-12x+16x$$

$$4x+9=9+4x$$

$$9=9$$

Infinite Solutions

$$\star \textcircled{6} 2-8(x+1)+6=3x-11x$$

$$2-8x-8+6=3x-11x$$

$$-8x=-8x$$

$$0=0$$

Infinite Solutions

$$\textcircled{7} 6(2x-1)=3(x+2)$$

$$12x-6=3x+6$$

$$9x=12$$

$$x=\frac{4}{3}$$

One Solution

$$\textcircled{8} 5x+1-2x+3=2x+x+4$$

$$3x+4=3x+4$$

$$4=4$$

Infinite Solutions

$$\textcircled{9} 2-4(5+5x)=3-20x$$

$$2-20-20x=3-20x$$

$$-18-20x=3-20x$$

$$-18=3$$

No Solution

Systems of Equations with No Solution or Infinite Solutions

Methods for Solving Systems of Equations

I) Substitution

II) Elimination

III) Graphing

Substitution Method

$$\textcircled{10} \begin{cases} x=y+2 \\ 3x+y=4 \end{cases}$$

$$3(y+2)+y=4$$

$$3y+6+y=4$$

$$4y+6=4$$

$$y=-\frac{1}{2}$$

$$x=-\frac{1}{2}+2$$

$$x=1.5 \text{ or } \frac{3}{2}$$

$$\textcircled{11} \begin{cases} x-2y=42 \\ 4x=y \end{cases}$$

$$x-2(4x)=42$$

$$x-8x=42$$

$$-7x=42$$

$$x=-6$$

$$4(-6)=y$$

$$-24=y$$

$$\textcircled{12} \begin{cases} x=2y-1 \\ 3y-4x=5 \end{cases}$$

$$3y-4(2y-1)=5$$

$$3y-8y+4=5$$

$$-5y+4=5$$

$$y=-\frac{1}{5}$$

$$x=2(-\frac{1}{5})-1$$

$$x=-\frac{7}{5}$$

Elimination Method

$$\textcircled{13} \begin{cases} 5x + 4y = 3 \\ 3x - 4y = -11 \end{cases}$$

$$\hline 8x = -8$$

$$\boxed{x = -1}$$

$$\rightarrow 3(-1) - 4y = -11$$

$$-3 - 4y = -11$$

$$\boxed{y = 2}$$

$$\textcircled{14} \begin{cases} 7x + 2y = 20 \\ 7x - 3y = 5 \end{cases}$$

$$\hline 5y = 15$$

$$\boxed{y = 3}$$

$$\rightarrow 7x + 2(3) = 20$$

$$7x + 6 = 20$$

$$\boxed{x = 2}$$

$$\textcircled{15} \begin{cases} 10x - 8y = 30 \\ 7x + 8y = 21 \end{cases}$$

$$\hline 17x = 51$$

$$\boxed{x = 3}$$

$$\rightarrow 7(3) + 8y = 21$$

$$21 + 8y = 21$$

$$8y = 0$$

$$\boxed{y = 0}$$

$$\textcircled{16} \begin{cases} -3x - y = -4 \\ -3x + 2y = -1 \end{cases}$$

$$\hline -3y = -3$$

$$\boxed{y = 1}$$

$$\rightarrow -3x - 1 = -4$$

$$\boxed{x = 1}$$

$$\textcircled{17} \begin{cases} -2x + 3y = -1 \\ 8x - 5y = 11 \end{cases}$$

$$\begin{cases} 4(-2x + 3y) = -4 \quad (4) \\ 8x - 5y = 11 \end{cases}$$

$$\hline \begin{cases} -8x + 12y = -4 \\ 8x - 5y = 11 \end{cases}$$

$$7y = 7$$

$$\boxed{y = 1}$$

$$\rightarrow -2x + 3(1) = -1$$

$$\boxed{x = 2}$$

$$\textcircled{18} \begin{cases} 4x + 5y = 9 \\ 3x + 10y = 13 \end{cases}$$

$$\begin{cases} 2(4x + 5y) = 9 \quad (2) \\ 3x + 10y = 13 \end{cases}$$

$$\hline \begin{cases} 8x + 10y = 18 \\ 3x + 10y = 13 \end{cases}$$

$$5x = 5$$

$$\boxed{x = 1}$$

$$4(1) + 5y = 9$$

$$\boxed{y = 1}$$

$$\textcircled{19} \begin{cases} -6x + 5y = -25 \\ 4x + 7y = 27 \end{cases}$$

$$\begin{cases} 2(-6x + 5y) = -25 \quad (2) \\ 3(4x + 7y) = 27 \quad (3) \end{cases}$$

$$\hline \begin{cases} -12x + 10y = -50 \\ 12x + 21y = 81 \end{cases}$$

$$31y = 31$$

$$\boxed{y = 1}$$

$$\rightarrow 4x + 7(1) = 27$$

$$\boxed{x = 5}$$

$$\textcircled{20} \begin{cases} 6x + 5y = -6 \\ 5x + 3y = -5 \end{cases}$$

$$\begin{cases} 3(6x + 5y) = -6 \quad (3) \\ 5(5x + 3y) = -5 \quad (5) \end{cases}$$

$$\hline \begin{cases} 18x + 15y = -18 \\ 25x + 15y = -25 \end{cases}$$

$$-7x = 7$$

$$\boxed{x = -1}$$

$$5(-1) + 3y = -5$$

$$\boxed{y = 0}$$

Forms of Linear Equations

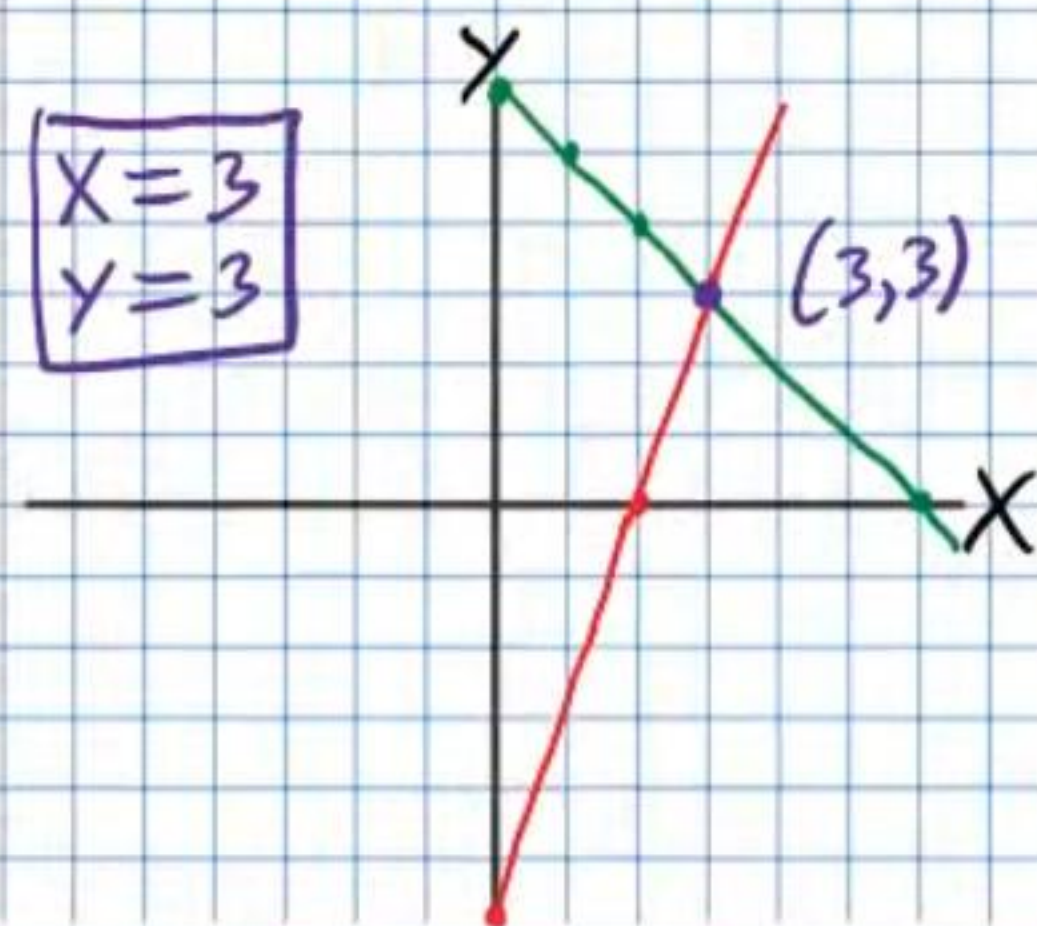
Standard Form
 $AX + BY = C$

Slope-Intercept Form
 $y = mx + b$

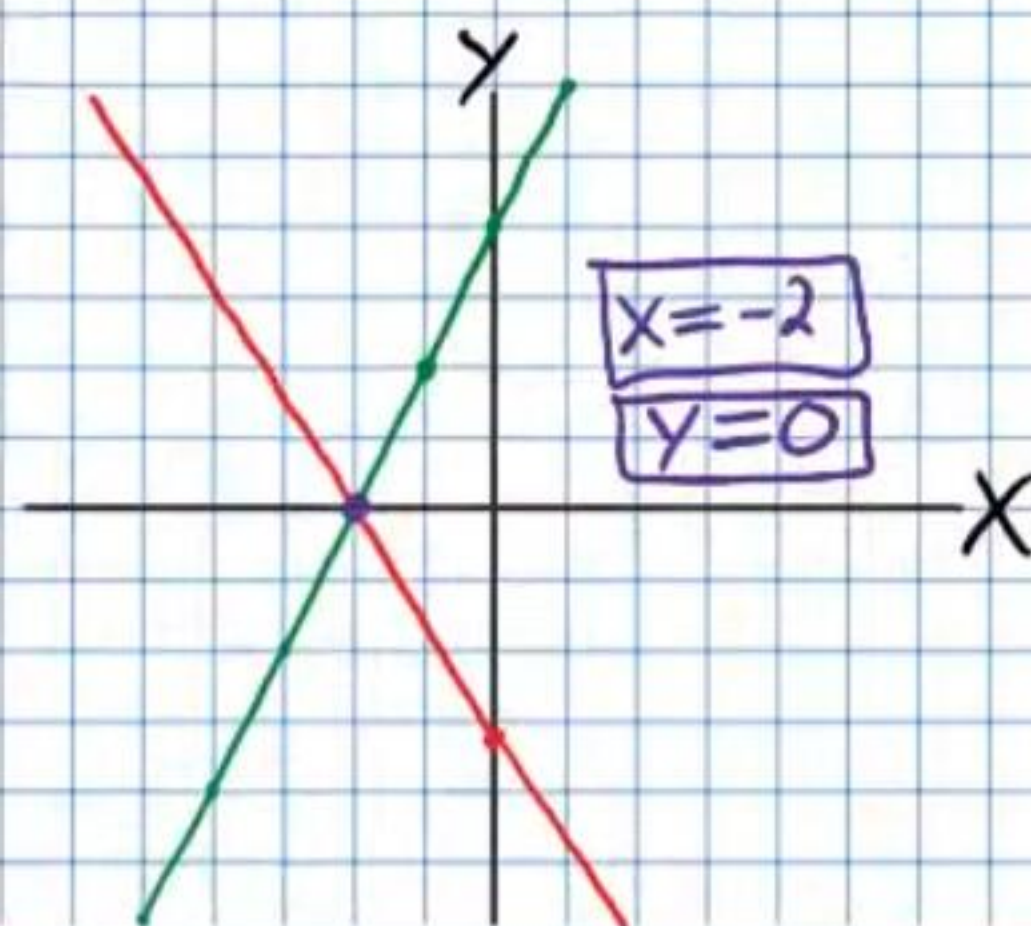
Point-Slope Form
 $y - y_p = m(x - x_p)$

Graphing Method

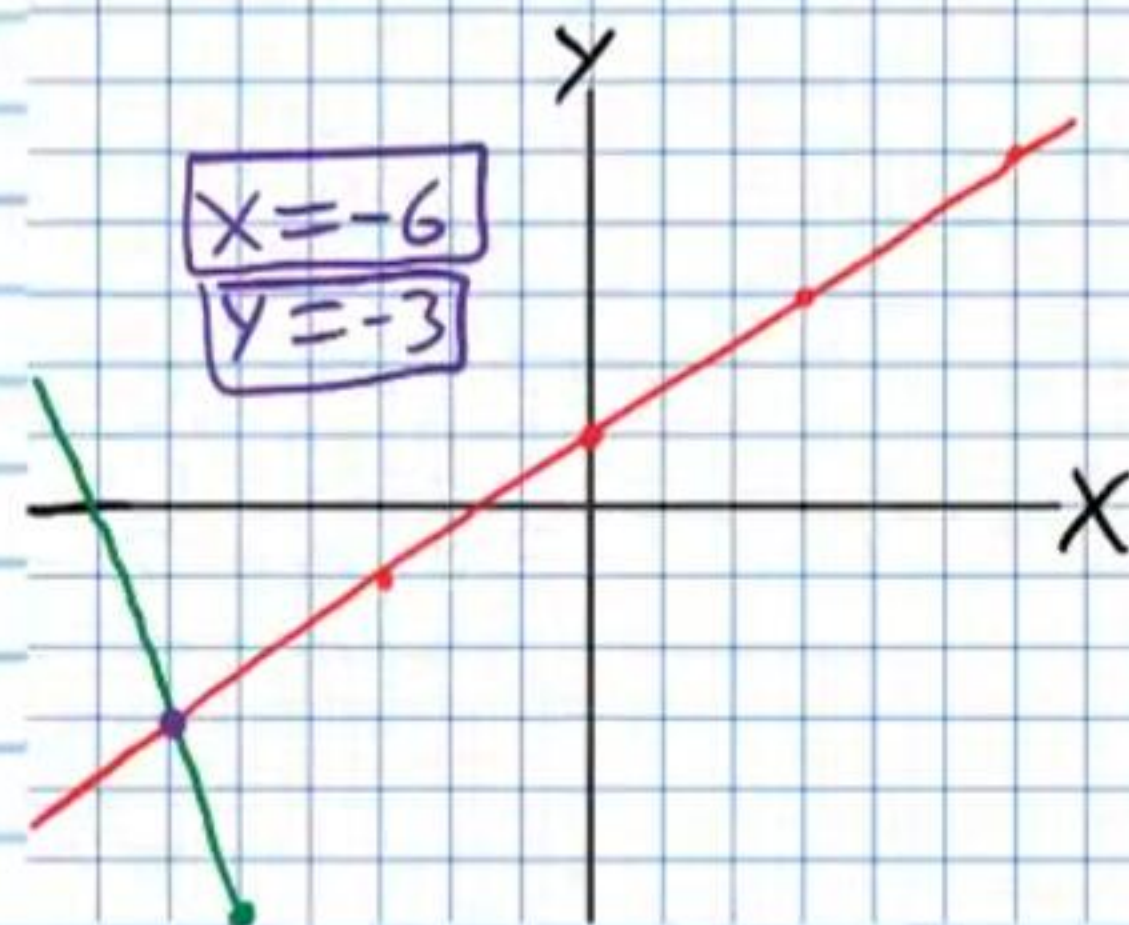
②① $\begin{cases} -3x + y = -6 \\ y = -x + 6 \end{cases}$



②② $\begin{cases} 5x + 3y = -10 \\ 2(x+1) = y - 2 \end{cases}$



②③ $\begin{cases} \frac{2}{3}x + 1 = y \\ y + 6 = -3(x+5) \end{cases}$



Determine if each of the following systems has one solution, no solution, or infinite solutions. Demonstrate algebraically.

②④ $\begin{cases} y = -5x + 3 \\ y = -5x + 10 \end{cases}$

$$-5x + 10 = -5x + 3$$

$$10 = 3$$

No Solution

②⑤ $\begin{cases} -2x + 5y = 4 \\ -2x + 5y = 1 \end{cases}$

$$0 = 3$$

No Solution

②⑥ $\begin{cases} x - 4 = y \\ 6 + y = x \end{cases}$

$$6 + y - 4 = y$$

$$2 + y = y$$

$$2 = 0$$

No Solution

②⑦ $\begin{cases} 4x + 2y = 1 \\ y = -2x + \frac{1}{2} \end{cases}$

$$4x + 2(-2x + \frac{1}{2}) = 1$$

$$4x - 4x + 1 = 1$$

$$1 = 1$$

Infinite Solutions

$$\textcircled{28} \begin{cases} 3x + 2y = -5 \\ -9x - 6y = 15 \end{cases}$$

$$\begin{cases} 3(3x + 2y) = -5(3) \\ -9x - 6y = 15 \end{cases}$$

$$\begin{cases} 9x + 6y = -15 \\ -9x - 6y = 15 \end{cases}$$

$$0 = 0$$

Infinite Solutions

Graphical Representations of Systems of Equations

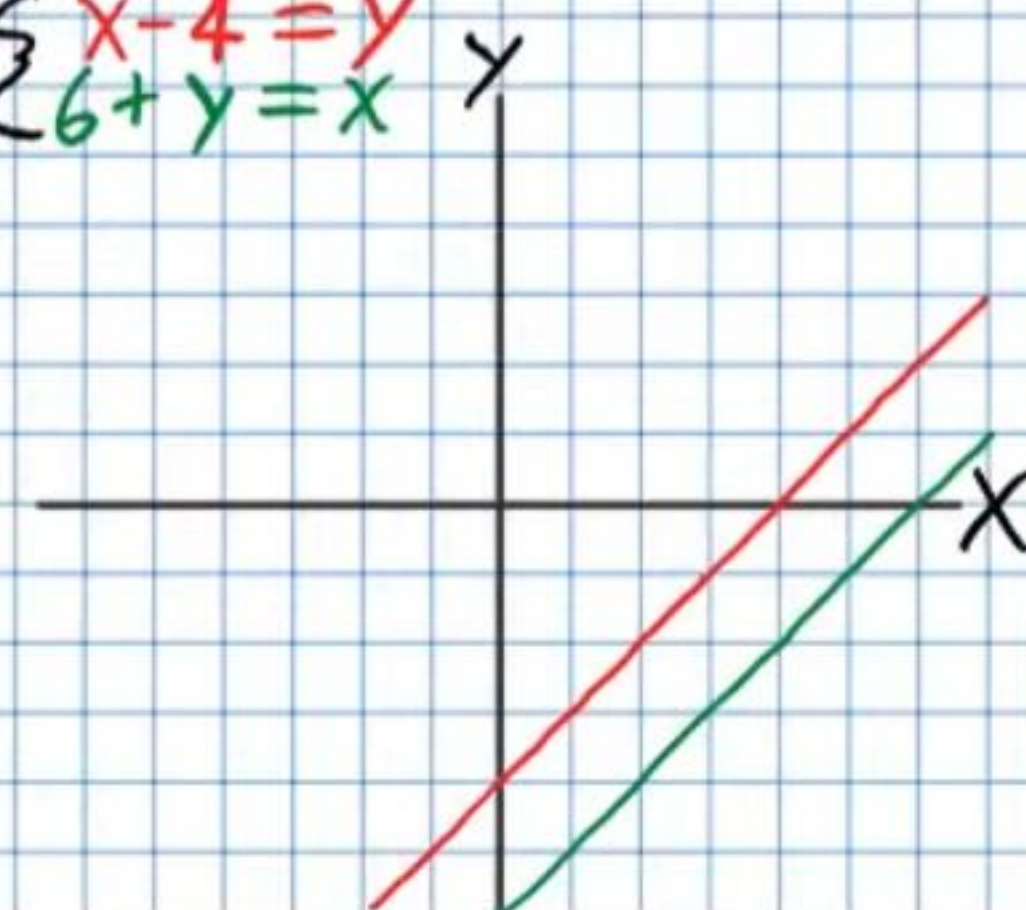
No Solutions

Infinite Solutions

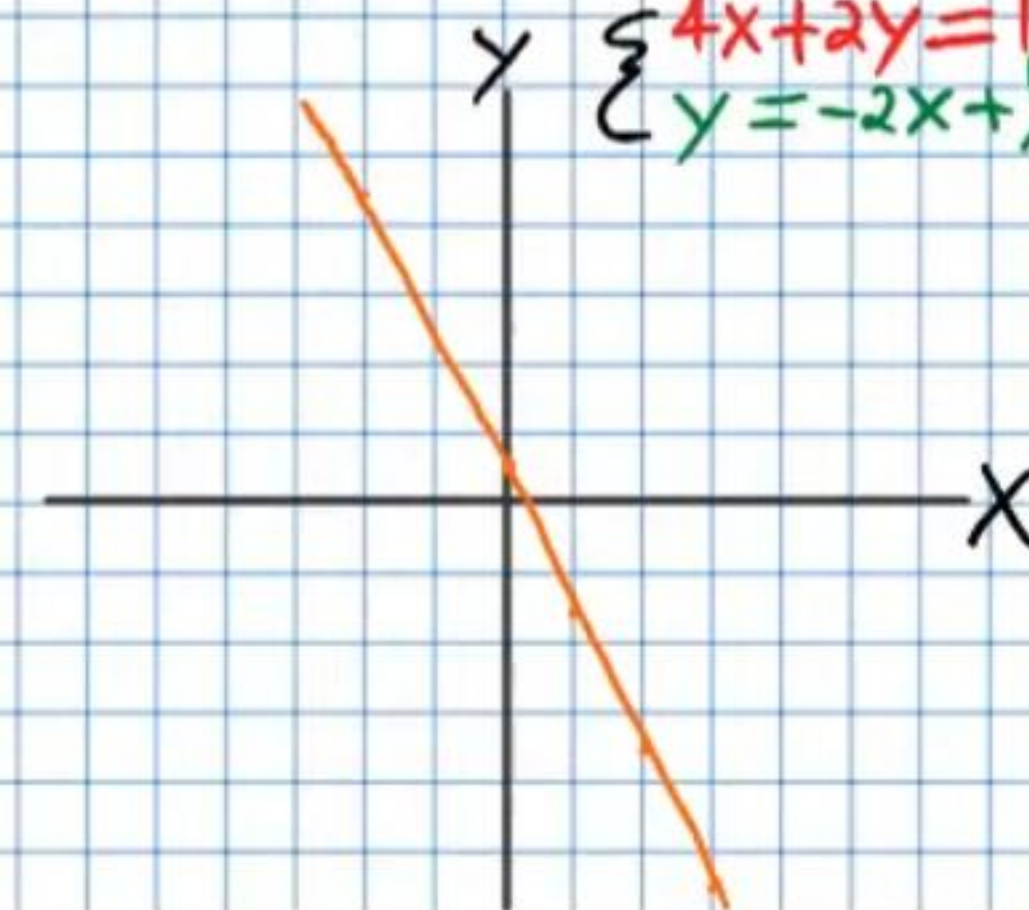
Separate Parallel Lines

Overlapping Parallel Lines

$$\begin{cases} x - 4 = y \\ 6 + y = x \end{cases}$$



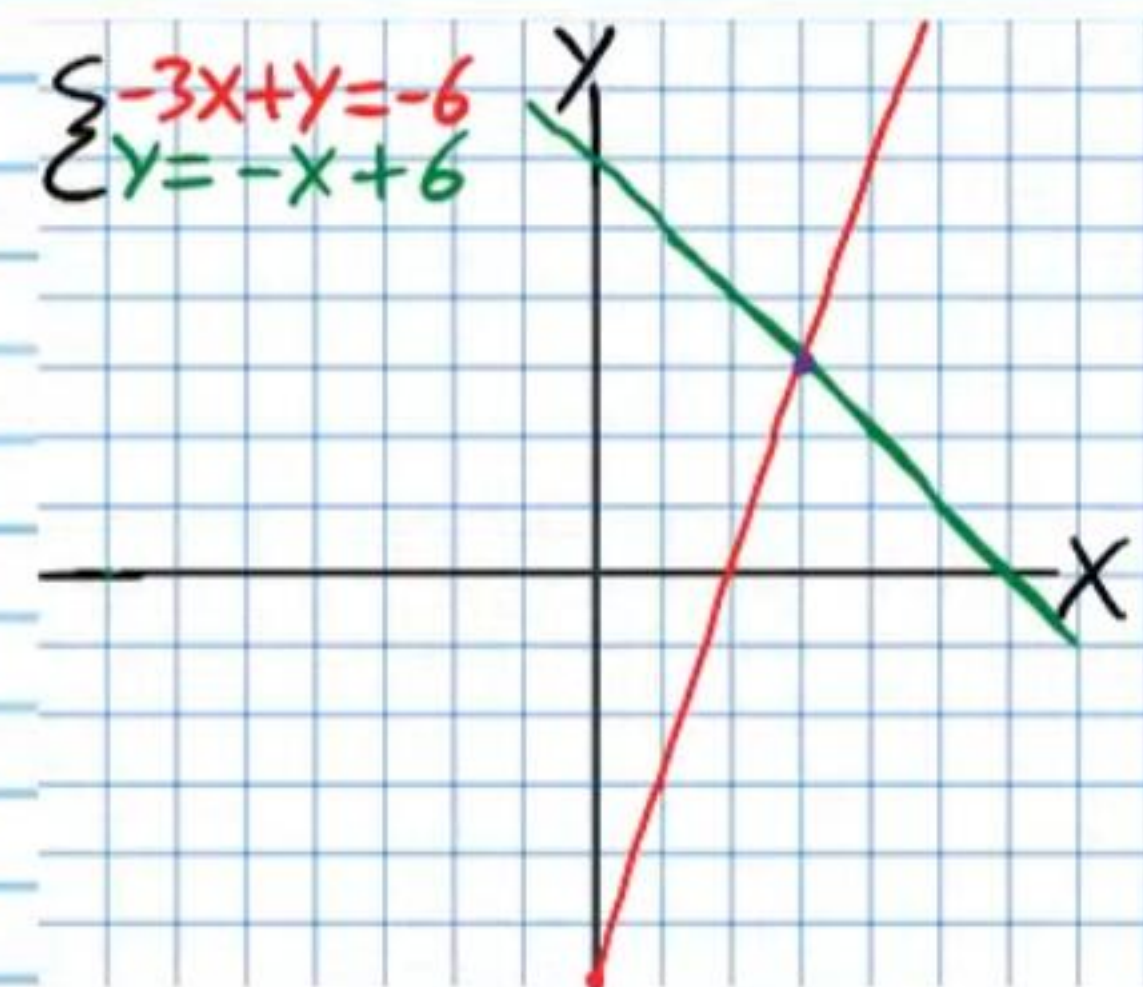
$$\begin{cases} 4x + 2y = 1 \\ y = -2x + \frac{1}{2} \end{cases}$$



One Solution

Lines Cross at One Point

$$\begin{cases} -3x + y = -6 \\ y = -x + 6 \end{cases}$$



Inequalities

Example i) $3 > 2$ Example ii) $-4 < -2$

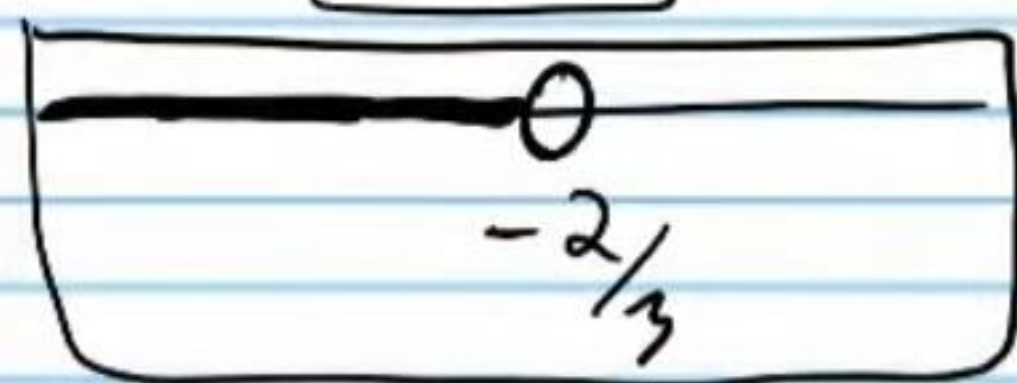
Find the algebraic solutions and the graphical solutions of the following inequalities.

②⑨ $2 + x > 4(x + 1)$

$$2 + x > 4x + 4$$

$$-2 > 3x$$

$$\boxed{-\frac{2}{3} > x}$$

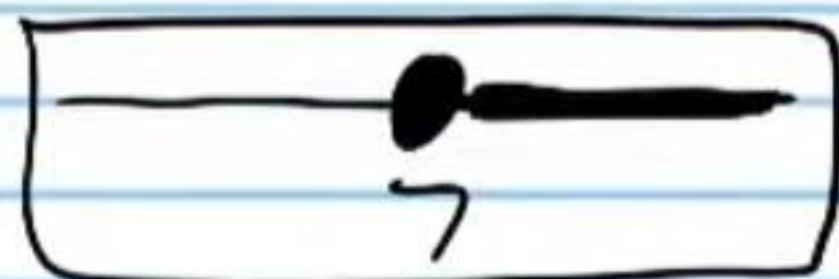


③① $-4 \geq -x + 3$

$$-7 \geq -x$$

$$-7 \geq (-1)x$$

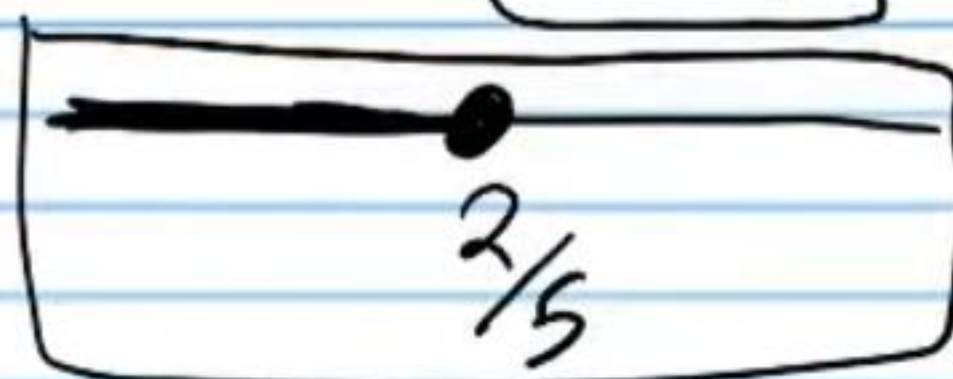
$$\boxed{7 \leq x}$$



Example ii) $-4 < -2$

③③ $5x + 2 \leq 4$

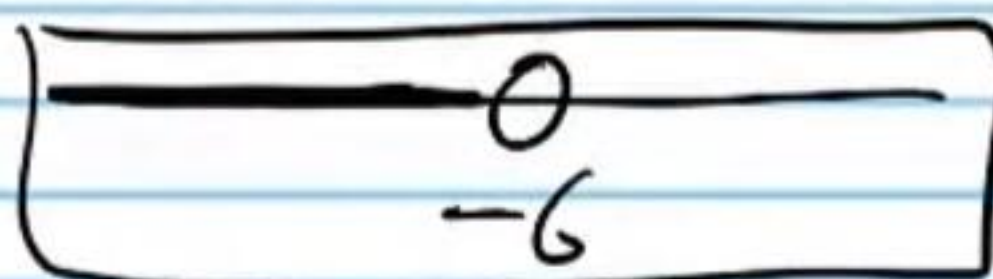
$$\boxed{x \leq \frac{2}{5}}$$



③② $6 < \frac{x}{-3} + 4$

$$2 < \frac{x}{-3}$$

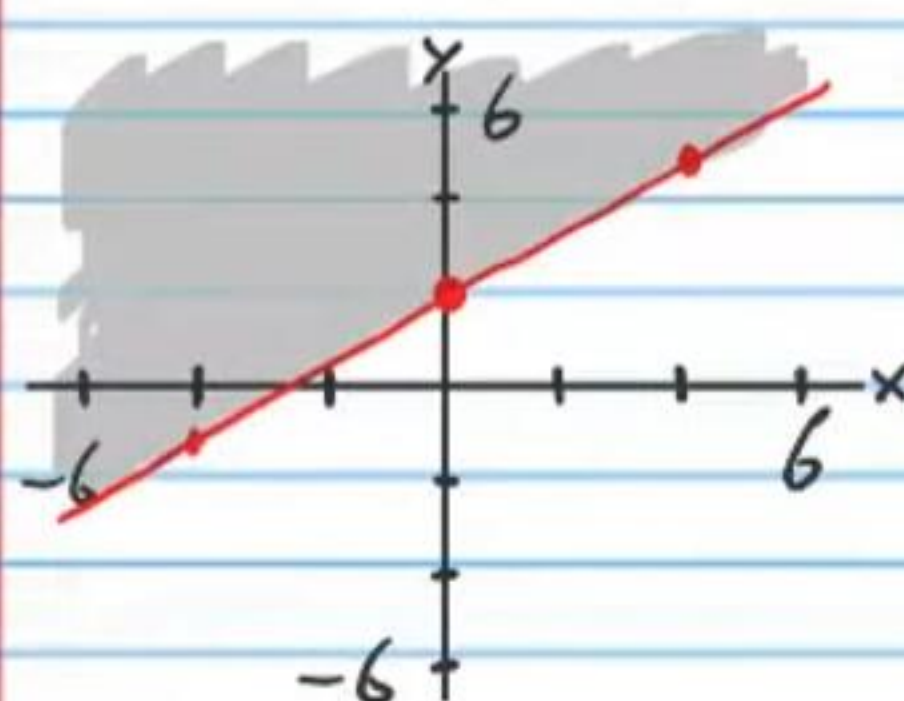
$$\boxed{-6 > x}$$



Graphing Linear Inequalities

Graph.

③③ $y \geq \frac{3}{4}x + 2$



Test Point
(0,0)

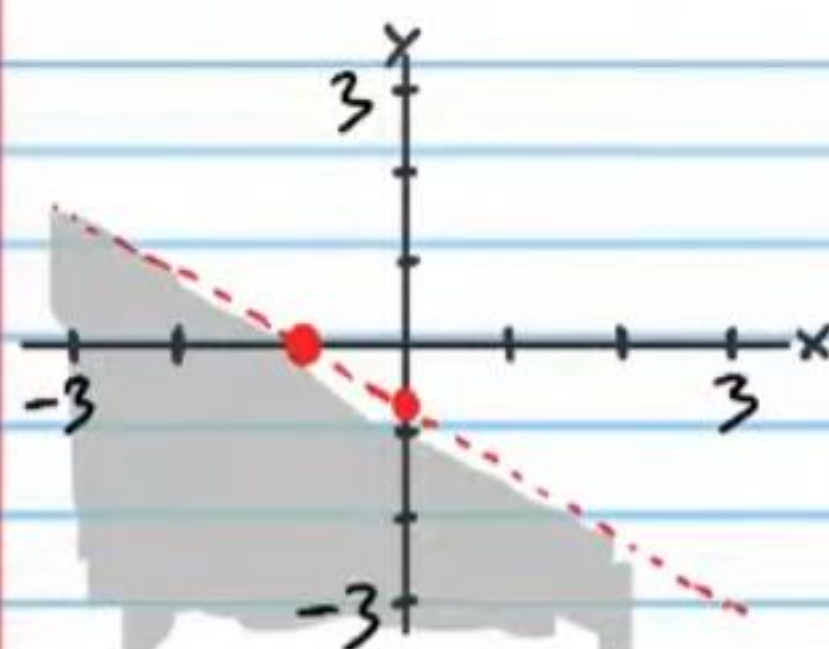
$$0 \geq \frac{3}{4}(0) + 2$$

$$0 \geq 0 + 2$$

$$0 \geq 2$$

✗

③④ $2x + 3y < -2$



Test Point
(0,0)

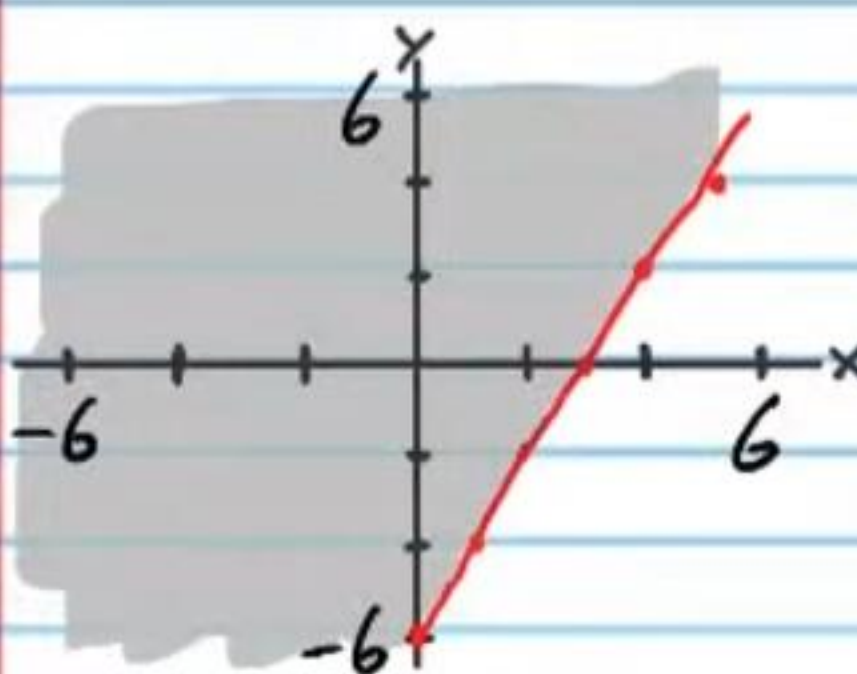
$$2(0) + 3(0) < -2$$

$$0 + 0 < -2$$

$$0 < -2$$

✗

③⑤ $2(x - 4) \leq y - 2$



Test Point
(0,0)

$$2(0 - 4) \leq 0 - 2$$

$$2(-4) \leq -2$$

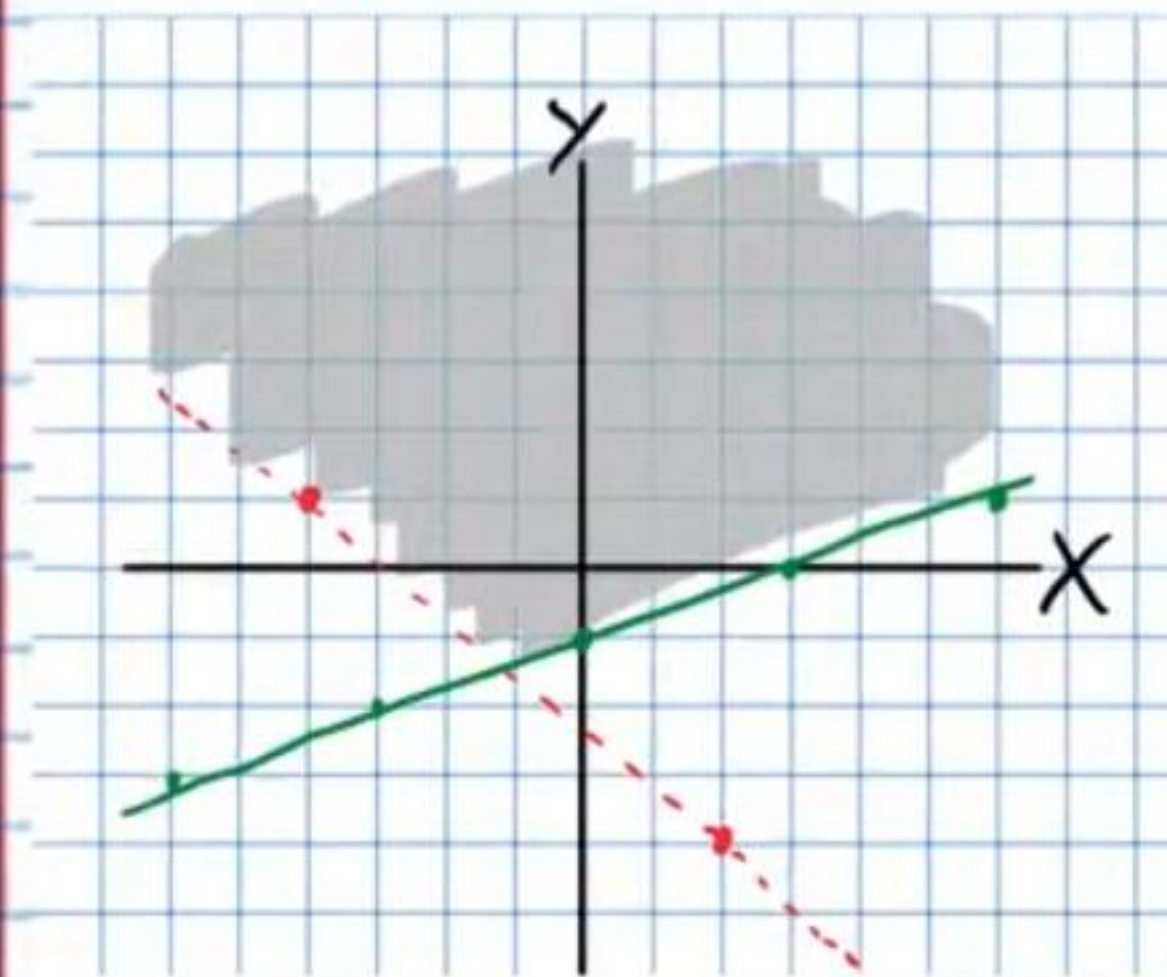
$$-8 \leq -2$$

✓

Graphing Systems of Linear Inequalities

Graph.

$$\textcircled{36} \begin{cases} y+4 > -\frac{5}{6}(x-2) \\ \frac{x}{3}-1 \leq y \end{cases}$$



Test Point 1
(0, -5)

$$-5+4 > -\frac{5}{6}(0-2)$$

$$-1 > -\frac{5}{6}(-2)$$

$$-1 > \frac{10}{6}$$

No

Test Point 2
(6, 0)

$$0+4 > -\frac{5}{6}(6-2) \quad \frac{6}{3}-1 \leq 0$$

$$4 > -\frac{5}{6}(4) \quad 2-1 \leq 0$$

$$4 > -\frac{20}{6} \quad 1 \leq 0$$

Yes

No

Test Point 3
(0, 0)

$$0+4 > -\frac{5}{6}(0-2) \quad \frac{0}{3}-1 \leq 0$$

$$4 > -\frac{5}{6}(-2) \quad -1 \leq 0$$

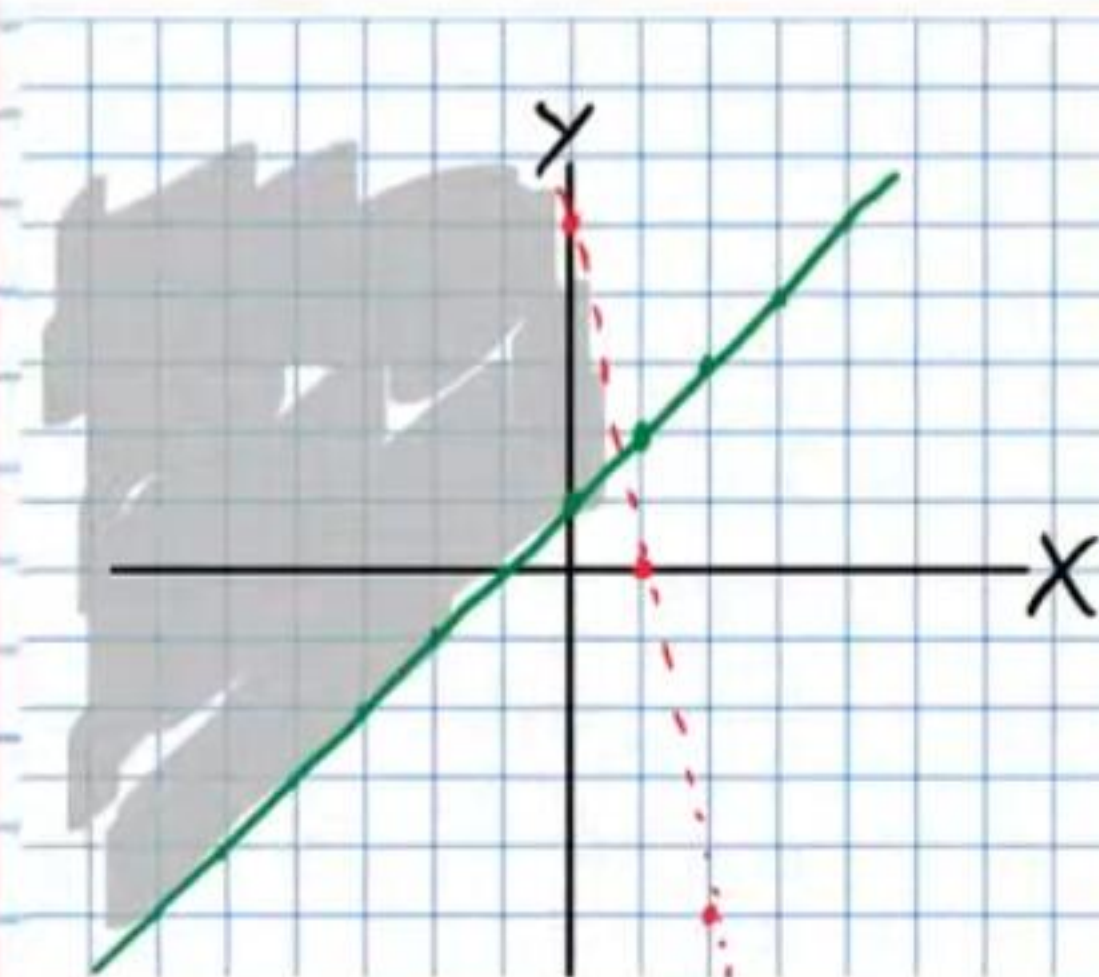
$$4 > \frac{10}{6} \quad \text{Yes}$$

$$4 > \frac{5}{3}$$

$$4 > 1\frac{2}{3}$$

Yes

$$\textcircled{37} \begin{cases} 5x+y < 5 \\ y \geq x+1 \end{cases}$$



Test Point
(5, 0)

$$5(5)+0 < 5$$

$$25 < 5$$

No

Test Point
(0, -2)

$$5(0)-2 < 5 \quad -2 \geq 0+1$$

$$-2 < 5 \quad -2 \geq 1$$

Yes

No

Test Point
(-5, 0)

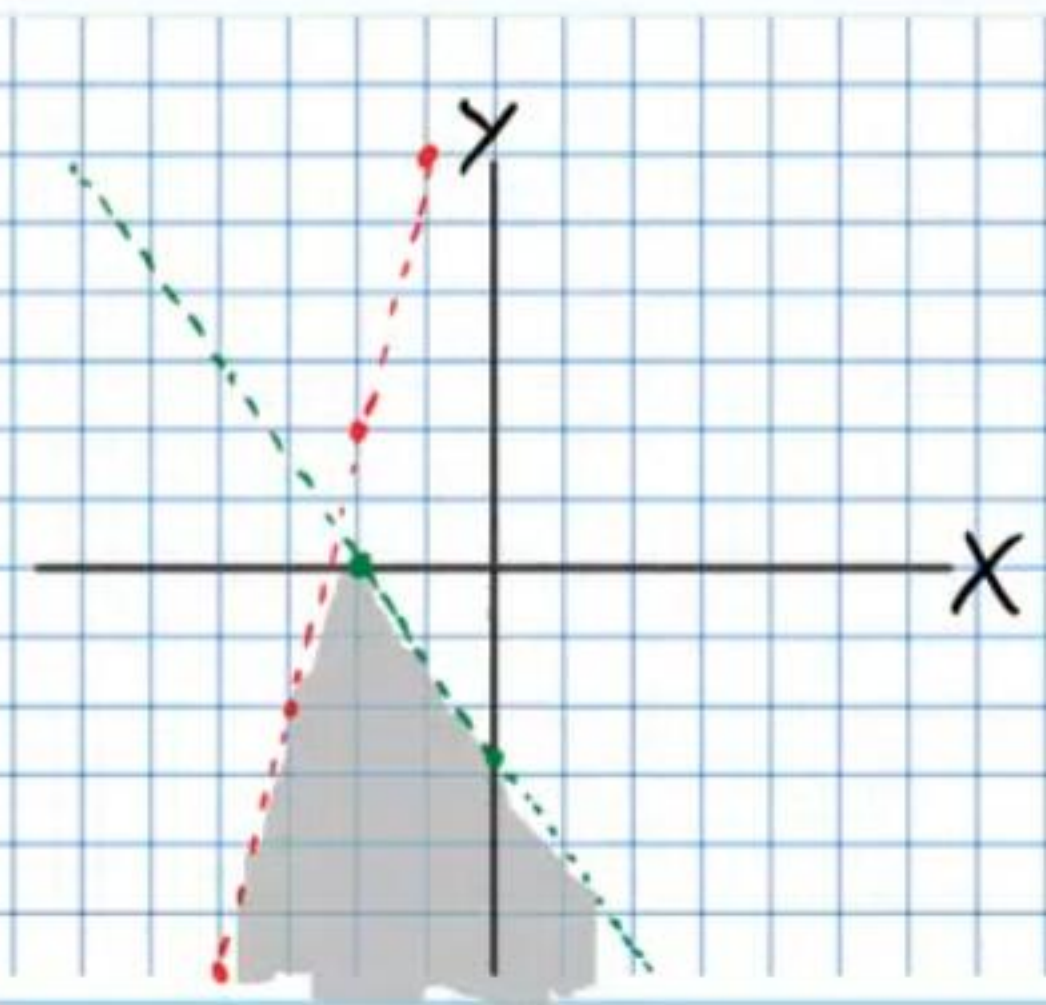
$$5(-5)+0 < 5 \quad 0 \geq -5+1$$

$$-25 < 5 \quad 0 \geq -4$$

Yes

Yes

$$\textcircled{38} \begin{cases} 4(x+2) > y-2 \\ -4x-3y > 8 \end{cases}$$



Test Point
(0, -10)

$$4(0+2) > -10-2 \quad -4(0)-3(-10) > 8$$

$$4(2) > -12 \quad 0+30 > 8$$

$$8 > -12 \quad 30 > 8$$

Yes

Yes